1 Mount Ijen in East Java is famous for its rare blue flames, visible at night. The phenomenon occurs when sulfur vapour, from the volcano's cracks burns, producing bright blue flames and sulfur dioxide. In the cool high-altitude air, some of the vapour condenses into solid sulfur.

(a) A team of environmental chemists were authorised to collect solid sulfur deposits near Mount Ijen's crater to investigate volcanic activity.

The chemists burnt the sulfur sample and measured the temperature change for a fixed amount of water placed in a calorimeter. The data from their experiment is shown in Table 1.1.

Table 1.1

mass of solid sulfur powder burnt /g	0.76
mass of water in beaker /g	150
initial temperature of water /^{o}C	29.8
final temperature of water / ^{o}C	39.7

(i) Construct an equation to represent the standard enthalpy change of combustion of solid sulfur, S.

..... [1]

(ii) Calculate the enthalpy change of combustion of solid sulfur, S, based on their experiment.

[2]

(iii) During volcanic activity, many sulfur-containing gases such as hydrogen sulfide, H_2S, and sulfur dioxide, SO_2, are released. In the atmosphere, H_2S can oxidise to form SO_2, and subsequently sulfur trioxide, SO_3, may be formed.

$$H_2S(g) + 3/2\ O_2(g) \rightarrow SO_2(g) + H_2O(l) \qquad \Delta H_r = -562\ \text{kJ mol}^{-1}$$

Use the data given in Table 1.2 to calculate the enthalpy change of combustion of solid sulfur, S.

Table 1.2

compound	ΔH_f / kJ mol^{-1}
$H_2S(g)$	–20.6
$H_2O(l)$	–285.8

[2]

(iv) Comment on the difference in values for the enthalpy change of combustion of solid sulfur determined in **(a)(ii)** and **(a)(iii)**.

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..... [1]

(v) With reference to a relevant chemical equation, explain how the release of sulfur-containing gases during volcanic activity can have a negative impact on the environment.

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..... [2]

(b) Sulfur in volcanic emissions primarily exists as a mixture of two stable isotopes: ^{32}S and ^{34}S.

The chemists analysed volcanic gas samples to determine the isotopes' relative abundance, which reveals the sulfur's origin.

- If the sample was enriched in ^{32}S, it originated from deep mantle degassing.
- If the sample was enriched in ^{34}S, it originated from hydrothermally recycled sources.

(i) 1.994 g of SO_2 evolved at Mount Ijen allowed the chemists to extract 1.00 g of elemental sulfur containing mixture of isotopes, ^{32}S and ^{34}S, for further analysis.

Calculate the percentage by mass of ^{32}S in the elemental sulfur extracted. You may use the chemical formula S to represent elemental sulfur in your calculations.

[3]

(ii) Hence, suggest the likely origin of the sulfur sample.

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..... [1]

While volcanic emissions affect the isotopic fractions of sulfur, the natural isotopic abundance of sulfur in Earth's environment is generally found to be present as follows.

Table 1.3

isotope	relative isotopic mass	percentage abundance / %
^{32}S	31.972	95.02
^{33}S	32.971	0.75
^{34}S	33.968	4.21
^{36}S	35.967	0.02

(iii) Use the data in Table 1.3 to calculate the relative atomic mass of sulfur, giving your answer to two decimal places.

[1]

[Total: 13]